



1. HARDWARE INVENTORY

M2 runs two node types: 2 rack-mounted Heltec LoRa V3 infrastructure nodes (ROUTER role) and 8 Heltec T114 field nodes (CLIENT role) loaned to attendees at events. Rack nodes bridge traffic to MQTT/Matrix via Pi 2. Field nodes are pre-configured checkout devices.

Infrastructure — Heltec LoRa V3 × 2 (ROUTER, rack-mounted)

Spec	Value
Chip	ESP32-S3 (Espressif)
Radio	SX1262 LoRa 915 MHz
GPS	None — infrastructure node, position not required
Display	0.96" OLED — 30s screen timeout
Power	5V USB-C via rack panel power strip
Firmware	Meshtastic — flash via esptool (.bin file)
Role	ROUTER — relays all mesh traffic, bridges to MQTT via Pi 2
Node names	M2 Community Node 01 / 02 short: M2-1, M2-2
Host	Pi 2 — USB serial /dev/ttyUSB0, /dev/ttyUSB1
MQTT bridge	mosquitto on Pi 2 — systemctl: meshtastic-mqtt

Field Nodes — Heltec T114 × 8 (CLIENT, loan devices)

Spec	Value
Chip	nRF52840 (Nordic Semiconductor)
Radio	SX1262 LoRa 915 MHz
GPS	L76K — built-in GPS for location sharing
Display	1.54" E-ink — always-on node info screen
Power	USB-C + Li-Po battery included in kit
Firmware	Meshtastic — flash via UF2 drag-and-drop (no esptool)
Role	CLIENT — personal loan device, one per attendee
Node names	M2 Field Node 01-08 short: MF-1 through MF-8
Config script	meshtastic/scripts/config_m2_t114.py
Channel	CommunityNode (shared PSK with rack nodes)

2. NETWORK ARCHITECTURE

M2 runs 2 rack-mounted Heltec V3 ROUTER nodes (infrastructure) plus 8 T114 CLIENT nodes (loan devices for attendees). The rack nodes connect to Pi 2 via USB serial. mosquitto bridges Meshtastic traffic to the internal MQTT broker, allowing Matrix and ATAK to receive mesh messages. T114 field nodes communicate over LoRa only — no serial connection to Pi 2.

```
Personal phone → Bluetooth → M2-1 or M2-2 (LoRa radio)
M2-1 / M2-2 → USB serial → Pi 2 /dev/ttyUSB0, /dev/ttyUSB1
Pi 2 mosquitto → MQTT topic → Matrix bridge / ATAK CoT relay
M2-1 <--> M2-2 (dual ROUTER — redundant coverage and relay)
```

MQTT bridge service

Service: meshtastic-mqtt (systemd, Pi 2)

Status: sudo systemctl status meshtastic-mqtt



3. SCRIPT REFERENCE

All scripts run from the project root directory.

Script (meshtastic/scripts/)	When	What it does
config_heltec_v3.py	Before event	Push PSK + config to 2 rack nodes. Run after PSK rotation.
config_m2_t114.py	Before event	Push PSK + config to 8 T114 field nodes. Run after PSK rotation.
gen_m2_channel_qr.py	Before event	Regenerate day-of QR card with current PSK.
generate_m2_mesh_handoff.py	When docs change	Rebuild this operator reference PDF.
generate_m2_mesh_runbook.py	When docs change	Rebuild the non-technical runbook PDF.
generate_m2_mesh_cheatsheet.py	When docs change	Rebuild the 1-page operator cheatsheet.
generate_m2_mesh_connection_card.py	When docs change	Rebuild T114 device checkout cards (8 nodes, 4 pages).

4. FIRST-TIME NODE SETUP

Flash and configure each node once. For pre-event PSK rotation, skip directly to Section 5 — only the config scripts need re-running.

T114 Field Nodes × 8 (UF2 drag-and-drop flash)

Step A — Download T114 firmware

Go to meshtastic.org/downloads. Select Heltec T114. Download the .uf2 file. No esptool required.

Step B — Enter bootloader mode on each T114

Double-tap the reset button quickly.

A USB drive appears on your computer. If no drive appears: hold BOOT while plugging in USB.

Step C — Copy firmware to the drive

Drag the .uf2 file onto the drive. The drive disappears and the node reboots automatically.

Step D — Configure all 8 T114 nodes

```
python meshtastic/scripts/config_m2_t114.py
```

Plug in each T114 via USB-C data cable when prompted. Script pushes: CommunityNode channel, PSK, US region, LONG_FAST, CLIENT role.

Heltec LoRa V3 × 2 (esptool flash — rack nodes only)

Step E — Flash via esptool

Download firmware for Heltec Wireless Tracker V3 from meshtastic.org/downloads (.bin file). Flash each rack node:

```
esptool.py --chip esp32s3 --port COM3 write_flash 0x0 firmware-heltec-v3-*.bin
# Replace COM3 with actual port (Device Manager → Ports)
```

Step F — Configure both rack nodes

```
python meshtastic/scripts/config_heltec_v3.py
```

Plug in each Heltec V3 via USB-C data cable when prompted. Script pushes: CommunityNode channel, PSK, US region, LONG_FAST, ROUTER role, timezone.



5. PRE-EVENT CHECKLIST

Run before every event. PSK rotation is mandatory — anyone who scanned the QR at a previous event still holds the old key.

PSK Rotation (required before every event)

Anyone who received the QR at a past event still holds the old key.

Rotate it every event to invalidate previous access.

Generate: `python -c "import os,base64; print(base64.b64encode(os.urandom(16)).decode())"`

Paste into CHANNEL_PSK_B64 in: `config_heltec_v3.py`, `config_m2_t114.py`, AND `gen_m2_channel_qr.py`

Save the new PSK to `M2_SECRETS.md` (z_Backups/ vault) immediately.

1. Generate new PSK and save.
2. Update CHANNEL_PSK_B64 in `config_heltec_v3.py`, `config_m2_t114.py`, `gen_m2_channel_qr.py`
3. Run `python meshtastic/scripts/config_heltec_v3.py` — plug in each Heltec V3
4. Run `python meshtastic/scripts/config_m2_t114.py` — plug in each T114
5. Run QR generator — prints to `meshtastic/pdf/M2_Mesh_ChannelQR_DayOf.pdf`
6. Print 2-up QR cards and cut — bring at least 4 cards to the event
7. Print T114 device checkout cards — `meshtastic/pdf/M2_Mesh_DeviceCard_Front.pdf`
8. Test on 2 phones: Bluetooth pair → scan QR → confirm CommunityNode visible
9. Verify MQTT bridge is running on Pi 2 before deployment

6. TROUBLESHOOTING

Problem	Fix
Bootloader drive not appearing after double-tap	Hold BOOT while plugging in USB to force bootloader. Try a different cable.
T114 stuck in bootloader (drive stays mounted)	Firmware copy may have failed. Re-copy the .uf2 file onto the drive.
T114 GPS not acquiring fix	Move outdoors. GPS cold start takes 2-5 min. E-ink shows GPS icon when locked.
No serial port after plugging in Heltec V3	Use a data-capable USB-C cable. Check Device Manager for port assignment.
<code>config_heltec_v3.py</code> finds no port	Install CH340 or CP2102 USB driver. Try a different USB port.
Node not appearing in Meshtastic app	Toggle phone Bluetooth off/on. Force-close app and reopen. Short-press node reset.
Nodes not routing messages to each other	Confirm both rack nodes show ROUTER role in app. Move closer. Check US region.
QR scan does nothing in app	Update Meshtastic app — older versions cannot import URL-based channels.
MQTT bridge offline	<code>ssh pi2 — sudo systemctl restart meshtastic-mqtt — check journalctl logs.</code>
Channel PSK mismatch — nodes can't decrypt	Re-run <code>config_heltec_v3.py</code> , <code>config_m2_t114.py</code> , and <code>gen_m2_channel_qr.py</code> .
Heltec V3 firmware needs update	Download latest .bin from meshtastic.org/downloads . Re-flash via esptool.
Node boots but no OLED display	Expected in some firmware builds — verify via Meshtastic app over Bluetooth.



7. REPLACING A NODE

If a node is lost or damaged, the replacement gets a new permanent node ID. Update documentation to keep the operator reference accurate.

Replacing a T114 Field Node (MF-1 through MF-8)

1. Flash firmware via UF2 (see Section 4, Steps A-C)
2. Run `config_m2_t114.py` — assign the same slot number as the replaced node
3. Verify new node appears in Meshtastic app with correct name (MF-x)
4. Update device checkout card label if node number changes

Replacing a Heltec V3 Rack Node (M2-1 or M2-2)

1. Flash firmware via esptool (see Section 4, Step E)
2. Connect via USB — run `config_heltec_v3.py` for the replacement slot
3. Verify new node in Meshtastic app and confirm it appears in the node list
4. Reconnect to Pi 2 USB and verify MQTT bridge picks up the new node ID
5. Rebuild this handoff PDF — `python meshtastic/scripts/generate_m2_mesh_handoff.py`